Assignment 2 (20% of total marks)

Due date: Saturday, 14 February 2026 by 9:00 pm (21:00 hours) Singapore time.

Scope

The tasks of this assignment cover the **data structure and algorithm**. Focusing on data structure and algorithm design. The assignment covers the topics discussed in **Topic 4**, and **Topic 5**.

The assignment consists of:

- **Part A**: Theory questions to assess your understanding of the subject matter and support the programming question's objective.
- **Part B**: Programming questions requiring the implementation of concepts related to algorithms and their complexity.

Marks for each question are indicated next to the respective question. The total mark for Assignment 1 is **100 marks**.

Marks Distribution

- **Part A**: 70%
- **Part B**: 30%
- **Total Marks**: 100
- **Weightage**: 15% of the total subject mark

Assessment Criteria

Marks will be awarded for:

- **Correctness**: Solutions must meet the requirements of the question.
- **Comprehensiveness**: Solutions should demonstrate a complete understanding of the topic.
- **Appropriate Application**: Answers should reflect proper application of the materials covered in this subject.

Deliverables

Part A: Theory Questions (Questions 1 to 3)

- Type your answers using **MS Word** and save the completed document as a **PDF file**.
- Alternatively, you may write your answers by hand onto a piece of paper or using tablet. Ensure your handwriting is **neat**, then scan or save your handwritten solutions into a **PDF file**.

Part B: Programming Question (Question 4)

- Implement your solution following the standard requirements provided for the question. You may use **C, C++, Java, or Python**.
- Execute your program and take a **screenshot of the output**.
- Save your source code in **pure ASCII text format** with the appropriate file extension (e.g., **.cpp** for C++, **.java** for Java, or **.py** for Python).
- Ensure your program is appropriately documented with comments.
- Provide clear **compilation instructions** in a readme.txt file. Include batch or makefiles if applicable for C or C++ programs.
    - *Note*: If no compilation instructions are provided, your lecturer will attempt compilation based on their environment. Any failure due to incompatibility will result in a "failure to compile" outcome.

Submission Instructions

- Combine the following into a single **ZIP file** named **SolutionA1.zip**:
    - Your solutions to the theory questions (in **PDF format**).
    - Your source code for the programming question (in **ASCII text format**).
    - The screenshot of your program’s output.
- Do not include subdirectories in your submission.

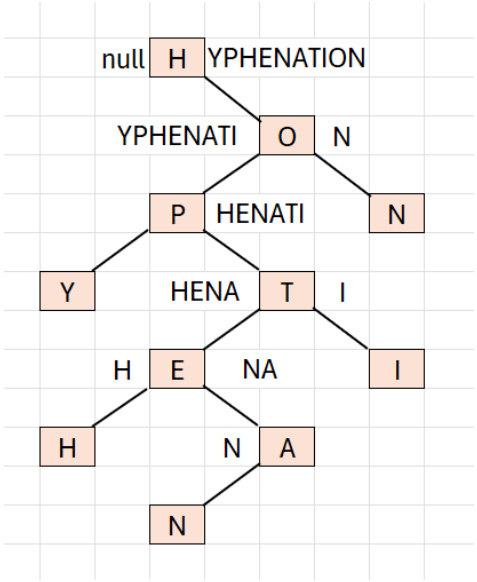
Part A: Theory Questions (70 Marks)

Question 1 – Binary Tree (15 Marks)

a) The INORDER traversal output of a binary tree is H,Y,P,H,E,N,A,T,I,O,N and the PREORDER traversal output of the same tree is H,O,P,Y,T,E,H,A,N,I,N. Construct the tree and determine the output of the POSTORDER traversal output.

(5.0 marks)

Sample solution includes but is not limited to the following:



Inorder: H, Y, P, H, E, N, A, T, I, O, N
Preorder: H, O, P, Y, T, E, H, A, N, I, N
Postorder: Y, H, N, A, E, I, T, P, N, O, H

b) One main difference between a binary search tree (BST) and an AVL (Adelson-Velski and Landis) tree is that an AVL tree has a balance condition, that is, for every node in the AVL tree, the height of the left and right subtrees differ by at most 1. Starting with an empty BST and AVL tree, insert the following values into the two trees (BST as well as AVL trees), in other words, first insert the values and construct a BST, then using the same set of the values insert and construct an AVL tree.

Tasks:

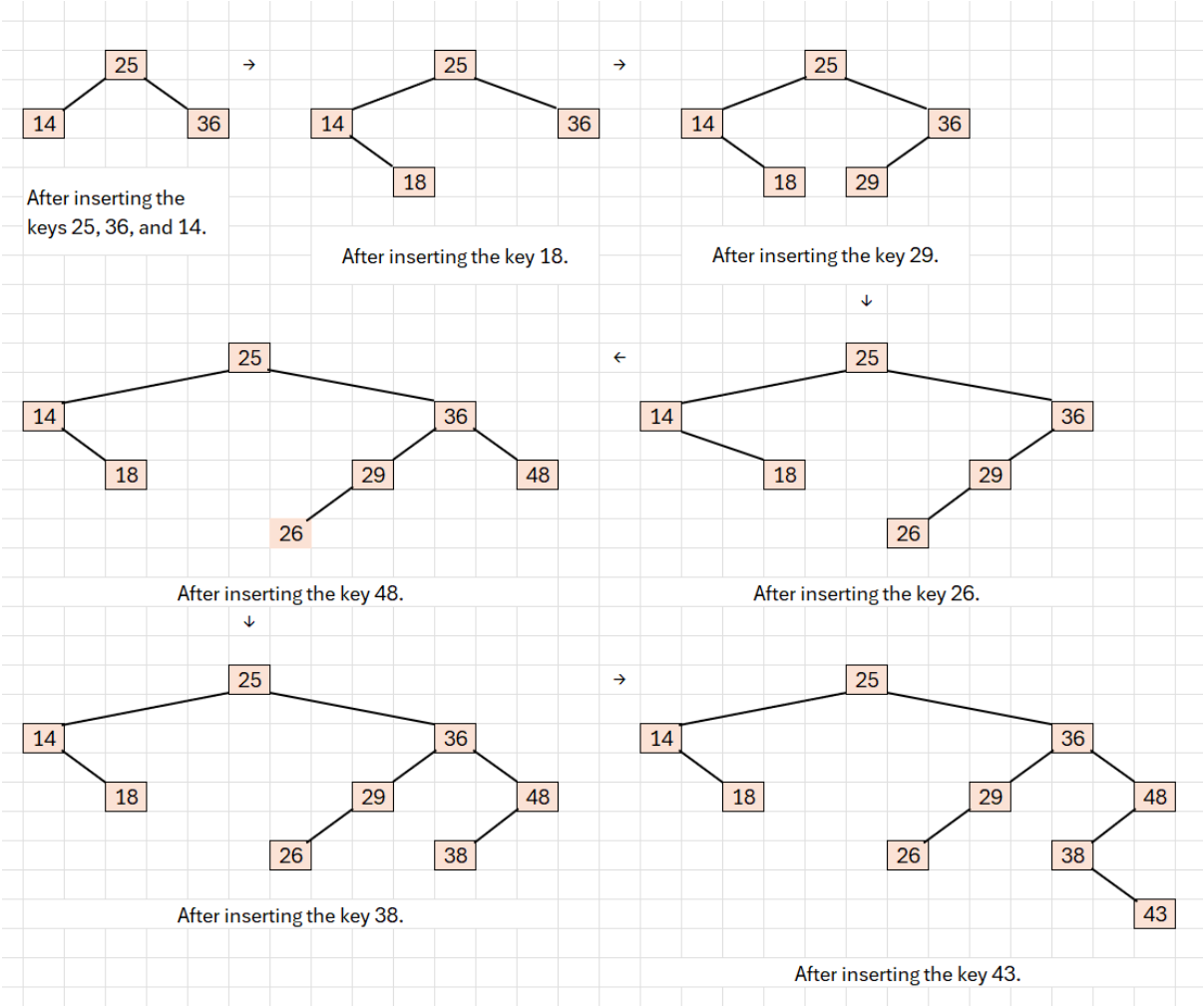
i. Starting with an empty BST (Binary Search Tree), insert the following values into the BST tree.

25, 36, 14, 18, 29, 26, 48, 38,43

If a key already existed, insert the new key to the LEFT sub-tree.

(2.0 marks)

Sample solution includes but is not limited to the following:



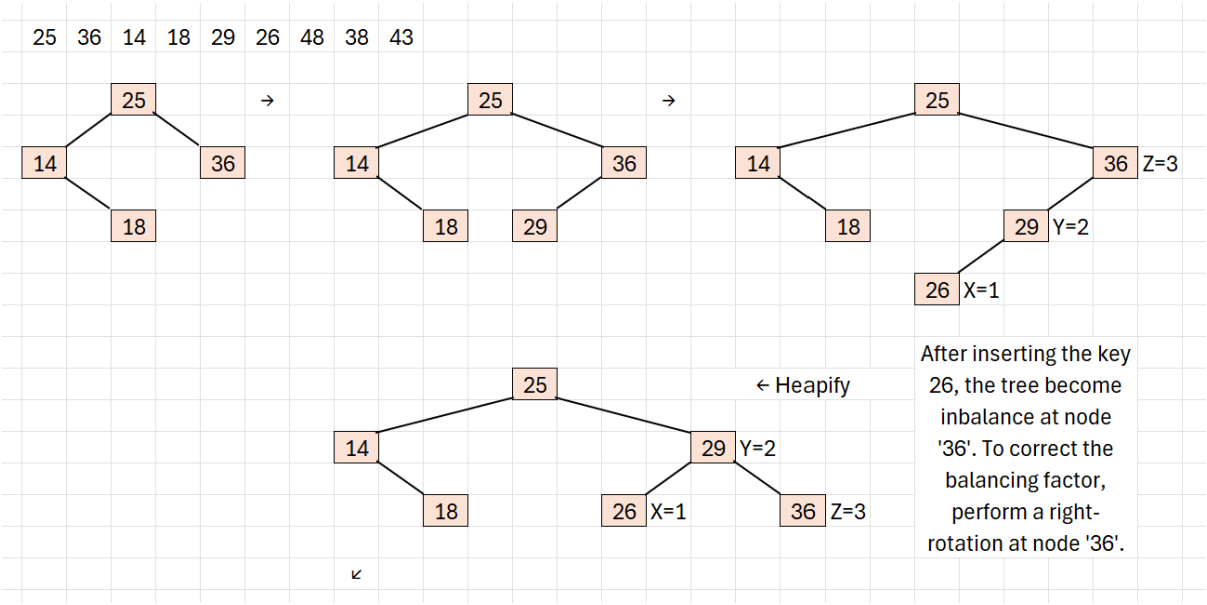
ii. Starting with an empty AVL (Balanced Binary Search Tree), insert the following values into the AVL tree.

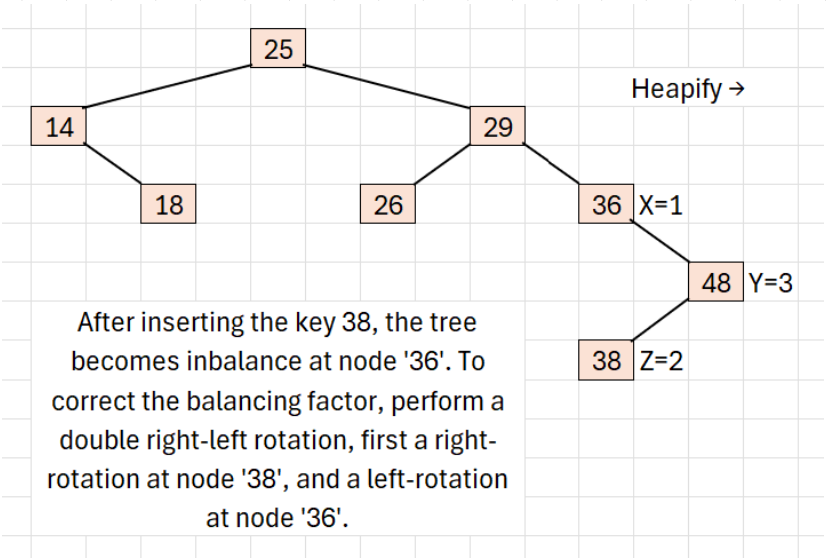
25, 36, 14, 18, 29, 26, 48, 38,43

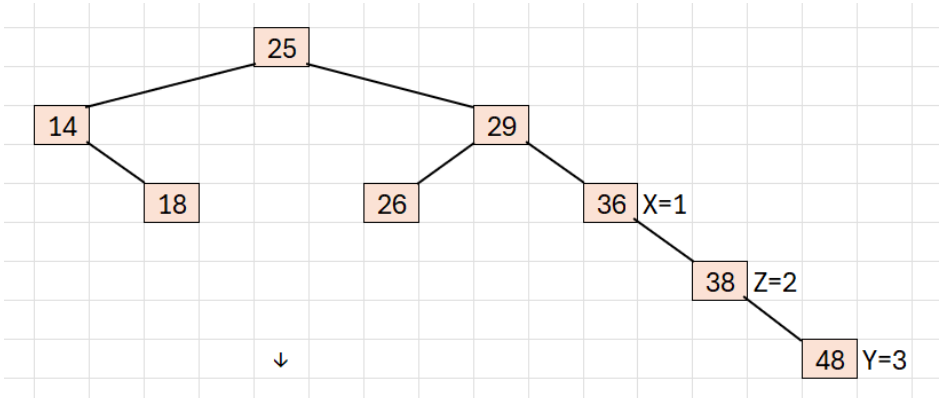
If a key already existed, insert the new key to the LEFT sub-tree. Show the rebalancing steps and rotations.

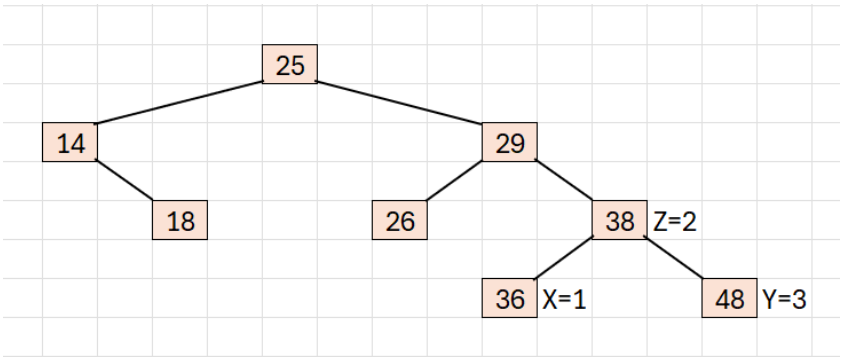
(3.0 marks)

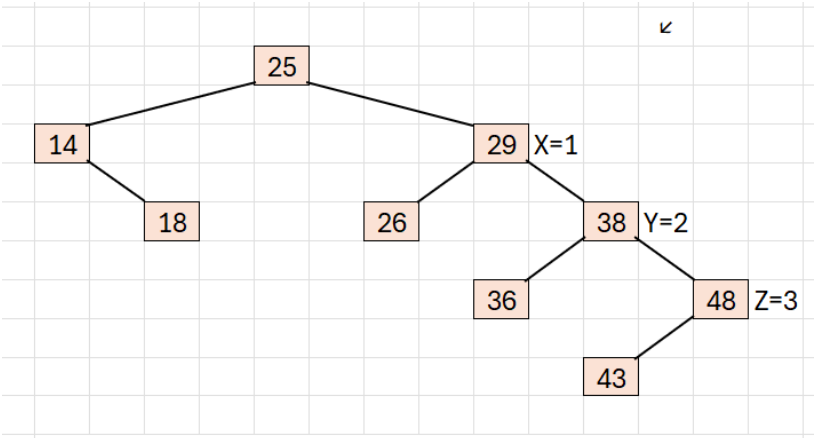
Sample solution includes but is not limited to the following:



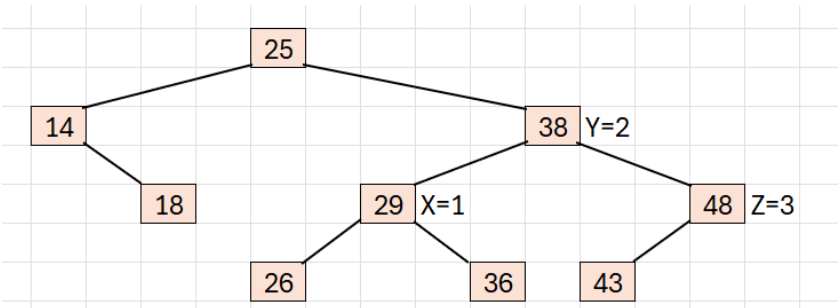






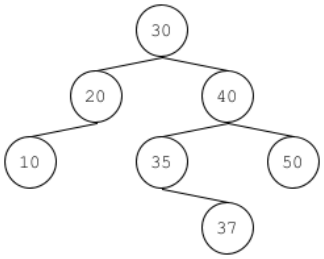


After inserting the key 43, the tree becomes inbalance at node '29'. To correct the balancing factor, perform a complex lef rotation at node '29'.



The construction of the AVL tree is completed.

c) Start with this initial BST (Binary Search Tree) tree:

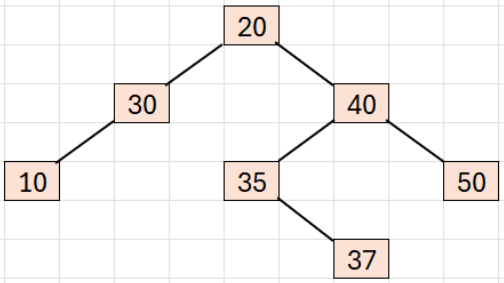


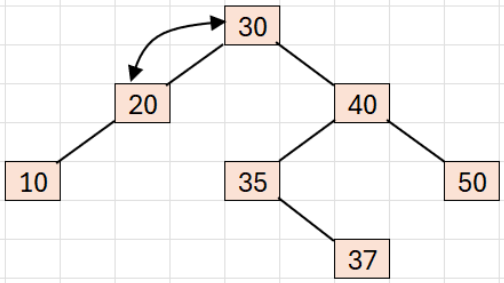
Task:

Delete 30 from the BST tree and draw the resulting tree. Explain the process of deletion.

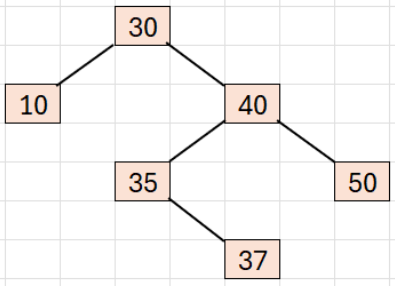
(2.0 marks)

Suggested solution includes but is not limited to the following:

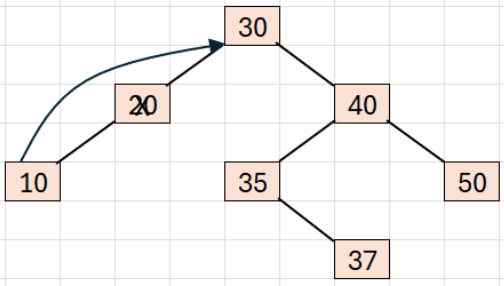




Swap the node '30' with node '20', the largest node value of the left subtree.

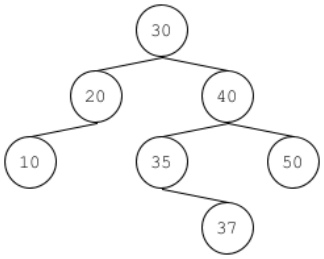


The node '20' is successfully deleted (removed) from the tree.



Delete (remove) the node '20' and attach the left subtree of node '20' to the left pointer of the parent node of node '20'.

d) Start with this initial AVL (Balanced Binary Search Tree) tree:

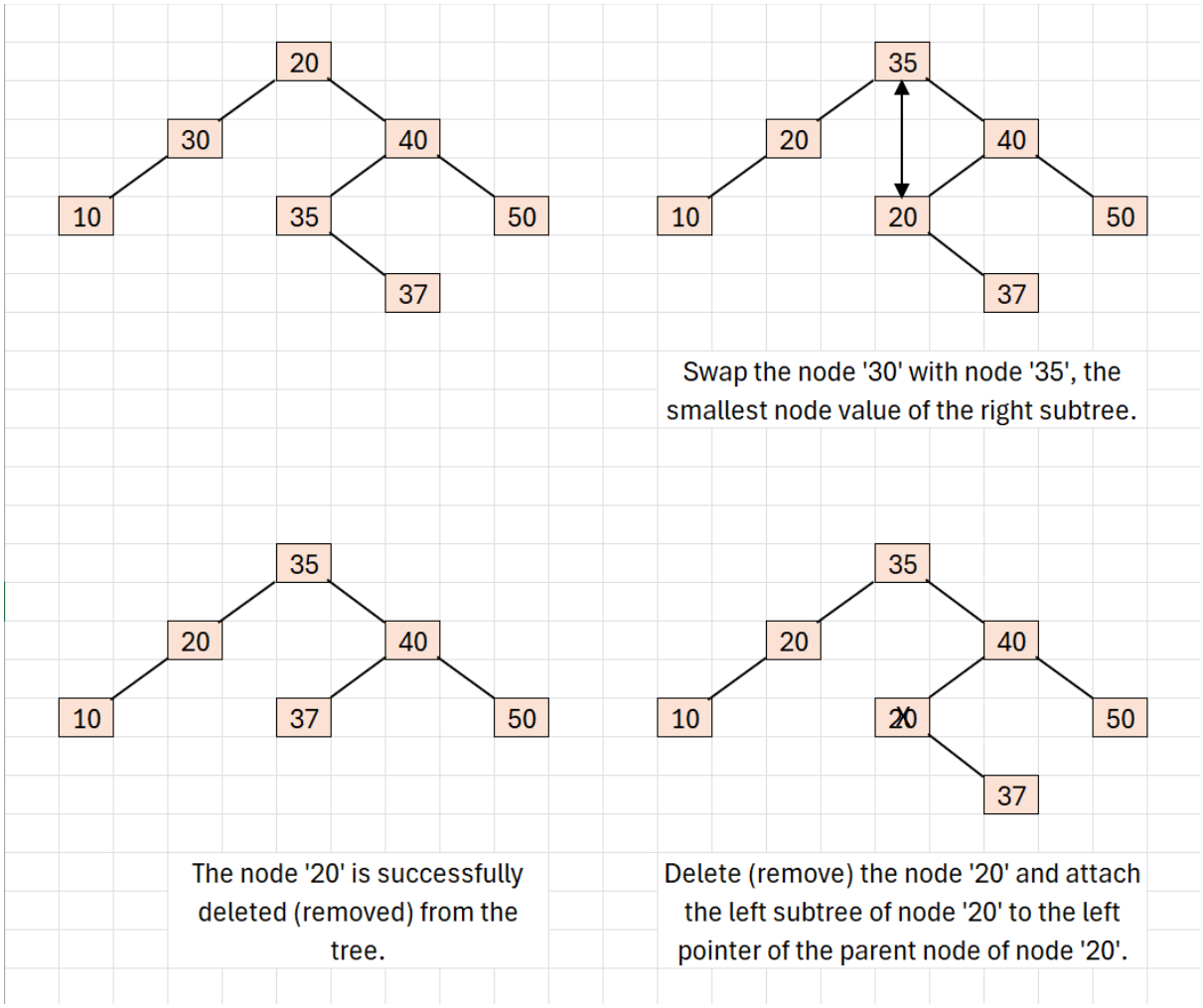


Task:

Delete 30 from the AVL tree. Show the rebalancing steps and rotations. Draw the AVL after the deletion.

(3.0 marks)

Suggested solution includes but is not limited to the following:



Question 2 – Queue and Stack (15 Marks)

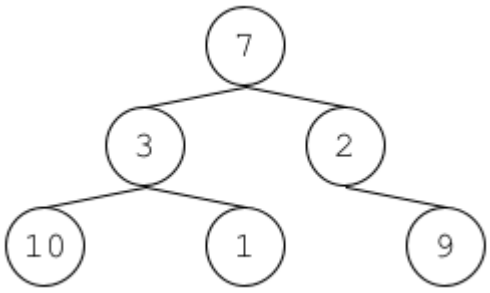
Two algorithms claim to return the same value from a binary tree.

```
Function B1(root)
    if (root == NULL) return -1
    x = root.data
    Q = new Queue()
    ENQUEUE(Q, root)

    while (!ISEMPTY(Q)) do
        t = DEQUEUE(Q)
        if (t.data > x) then
            x = t.data
        endIf
        ENQUEUE(Q, t.left)
        ENQUEUE(Q, t.right)
    endWhile

    return x
EndFunction
```

```
Function B2(root)
    if (root == NULL) return -1
    t = root
    while (t.right != NULL) do
        t = t.right
    endWhile
    return t.data
EndFunction
```

Note: the function ENQUEUE only inserts a new element in the queue if this element is not NULL. In other words, ENQUEUE ignores NULL.

a) For the following Binary Tree:



i. What is returned by B2(root)? **(2.5 marks)**

The function B2 will return the value 9, which is the leave node of the right most path of the root node 7.

ii. What is returned by B1(root)? **(2.5 marks)**

The function B1 will return the value 10, which is the largest value in the tree rooted at node 7.

b) For the Binary Tree in part (a), what is the content of the queue immediately before returning from the execution of function A1(root)? **(5.0 marks)**

The content of the queue immediately before returning from the execution is empty.

c) Re-write the function B2, in pseudocode, using recursive function calls. You **may not** use any form of iteration. **(5.0 marks)**

ALGORITHM 2 Recursive

```
Function A2R(root)
  if (root == NULL)
    return -1
  if(root->left == NULL)
    return root->data
  else
    A2R(root->left)
End function
```

Question 3 – Associative Table (20 Marks)

Consider a hash table of size 13 with hash function $h(x) = 3x \ mod \ 13$. Draw the table that results after inserting, in the given order, the following values: 18, 41, 22, 44, 59, 32, 31, 73, 14 for each of the three scenarios below:

a) When collisions are handled by separate chaining; **(5.0 marks)**

b) When collisions are handled by linear probing; **(5.0 marks)**

c) When collisions are handled by double hashing using a second hash function $h_2(x) = 13 - (x \ mod \ 13)$. Hint, the overall (combined) hash function is $H(x) = (\ h(x) + \ i \ \times \ h_2(x)\ ) \ mod \ 13$, where i = 0, 1, 2, 3, … **(5.0 marks)**

d) When collisions are handled by quadratic probing with a quadratic probe function $h_3(x, i) = (h(x) + 0.5\ i + 0.5\ i^2) \ mod \ 13$ where $i = 1, 2, 3, ….$ **(5.0 marks)**

Suggested solution included but not limited to:

a) Collisions are handled by separate chaining:

The list of keys: 18, 41, 22, 44, 59, 32, 31, 73, 14

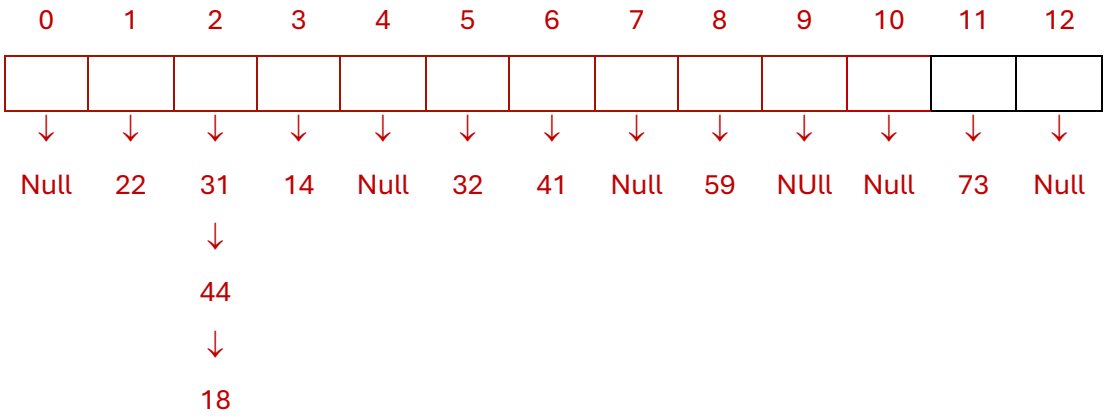
The hash function: $h(x) = 3x \ mod \ 13$

The hash address:

The hash addresses:

X	18	41	22	44	59	32	31	73	14
$h(x)$	2	6	1	2	8	5	2	11	3

The hash table using separate chaining collision resolution:



b) Collisions are handled by linear probing:

The list of keys: 18, 41, 22, 44, 59, 32, 31, 73, 14

The hash function: $h(x) = 3x \ mod \ 13$

The hash address:

The hash addresses:

X	18	41	22	44	59	32	31	73	14
$h(x)$	2	6	1	2	8	5	2	11	3

The hash table using linear probing collision resolution:

0	1	2	3	4	5	6	7	8	9	10	11	12
Null	22	18	44	31	32	41	14	59			73	

c) Collisions are handled by double hashing:

The list of keys: 18, 41, 22, 44, 59, 32, 31, 73, 14

The hash function: $h(x) = 3x \ mod \ 13$

The secondary hash function: $h_2(x) = 13 - \ (x \ mod \ 13)$

The double hash function: $H(x) = \big(h(x) + \ i \ \times \ h_2(x)\big) \ mod \ 13, \ \ where \ i = 0, 1, 2, 3, \ldots$

The hash address:

The hash addresses:

X	18	41	22	44	59	32	31	73	14
$h(x)$	2	6	1	2	8	5	2	11	3
$h_2(x)$				10			10, 5, 0		
$H(x)$	2	6	1	10	8	5	0	11	3

The hash table using double hashing collision resolution:

0	1	2	3	4	5	6	7	8	9	10	11	12
31	22	18	14		32	41		59		44	73	

d) Collisions are handled by quadratic probing:

The list of keys: 18, 41, 22, 44, 59, 32, 31, 73, 14

The hash function: $h(x) = 3x \ mod \ 13$

The quadratic function: $h_3(x, i) = (h(x) + 0.5\ i + 0.5\ i^2) \ mod \ 13$

The hash address:

The hash addresses:

X	18	41	22	44	59	32	31	73	14
h(x)	2	6	1	2	8	5	2	11	3
$h_3(x, 1)$	3	7	2	3	9	6	3	12	4
$h_3(x, 2)$				5			5		
$h_3(x, 3)$	3	7	2	5	9	6	8	12	4

The hash table using linear probing collision resolution:

0	1	2	3	4	5	6	7	8	9	10	11	12
Null	Null	22	18	14	44	32	41	31	59	Null	Null	73

Question 4 – Heap (20.0 marks)

a) Given a binary min-heap h stored as an array of n keys, design an algorithm $convertToMaxHeap(h)$ that transforms h **in-place** into a valid binary max-heap. Your algorithm must run in $\boldsymbol{O(n\ log\ n)}$ time. Provide a brief runtime analysis to justify the bound.

(10.0 marks)

Sample solution including but not limited to the following:

```
convertToMinHeap( h )
BEGIN
    SET last TO length(h)
    SET nlast TO (last−1 / 2)   // approximately half of the heap size
    REPEAT
        heapify(h, nlast)
        DECREMENT nlast BY 1
    UNTIL nlast = 0                    // reaches the root
END convertToMinHeap
```

```
heapify(h, i)
Begin
    SET last TO length(h)
    SET k TO i
    REPEAT
        SET j TO k
        IF ( 2j+1 <= last ) AND ( h[2j+1] < h[k] )
            SET k TO 2j+1
        ENDIF
        IF ( (2j + 2) <= last ) AND ( h[2j+2] < h[k] )
            SET k to 2j + 2
        ENDIF
```

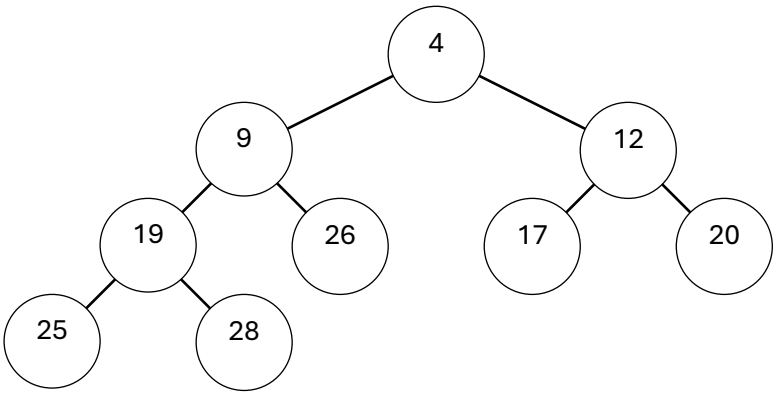
Swap(h[j], h[k])

UNTIL j = k

END heapify

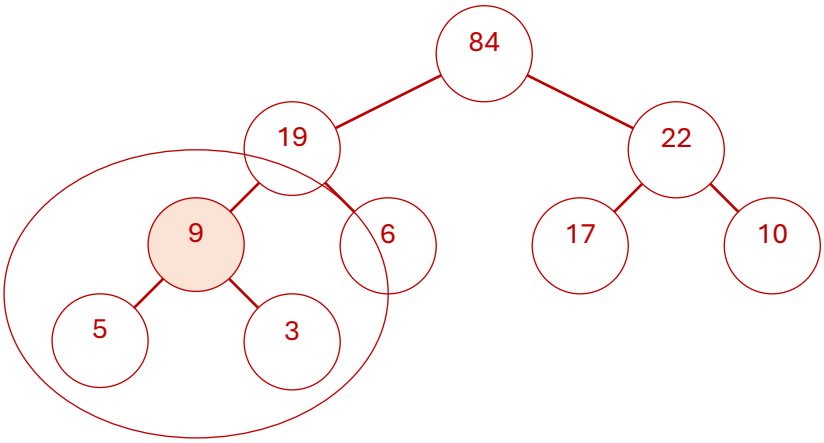
Starting from $nlast$ (the parent of the last element of a heap), the algorithm runs $heapify$ until $nlast$ reaches the root. This process calls $\frac{1}{2}n$ time of $heapify$, where n is the total number of items in the heap. When $heapify$ is called, starting from i the value will have to be exchanged down (if it violates the minimum heap property) with its child at every level of the heap, until it reaches the lowest level. In its worst case, the number of exchanges done will be equal to the height of the heap tree, which is $\lg n$. Hence, the worst case running time is upper bounded by $O(nlgn)$.

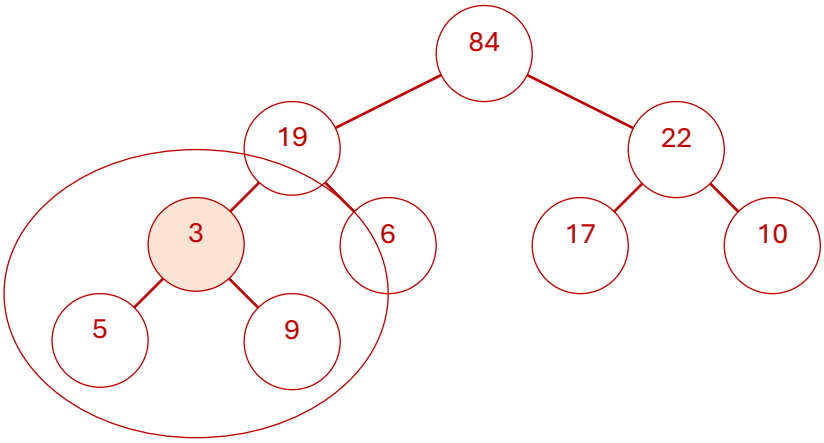
a. Given the following maximum heap, illustrate the process of converting it into a minimum heap using your algorithm described in (a). You need to show the intermediate processes. **(10.0 marks)**

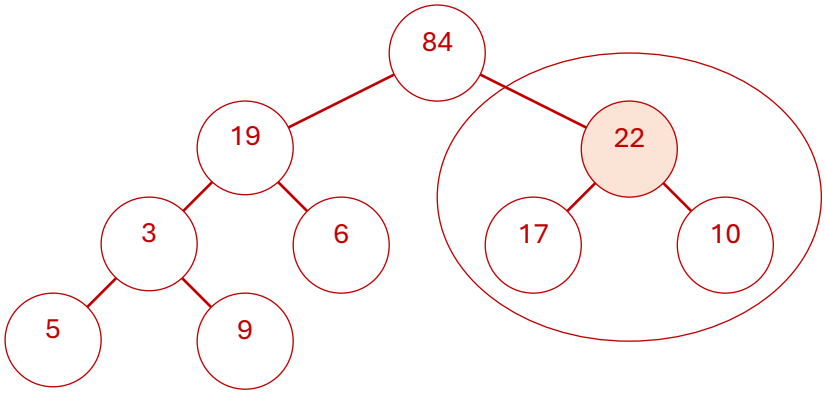


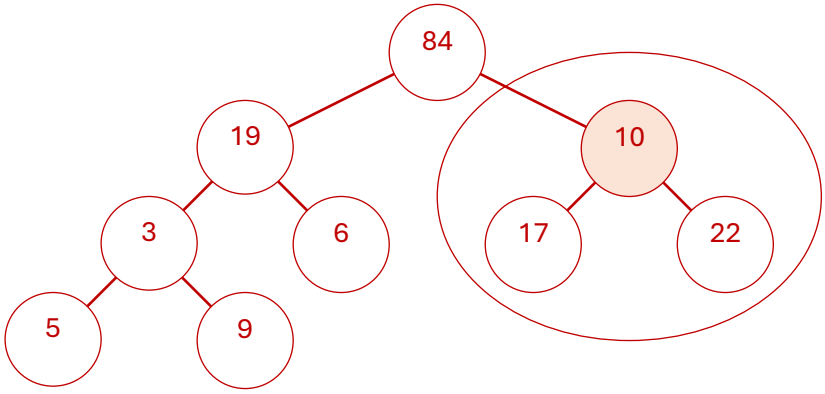
Sample solution including but not limited to the following:

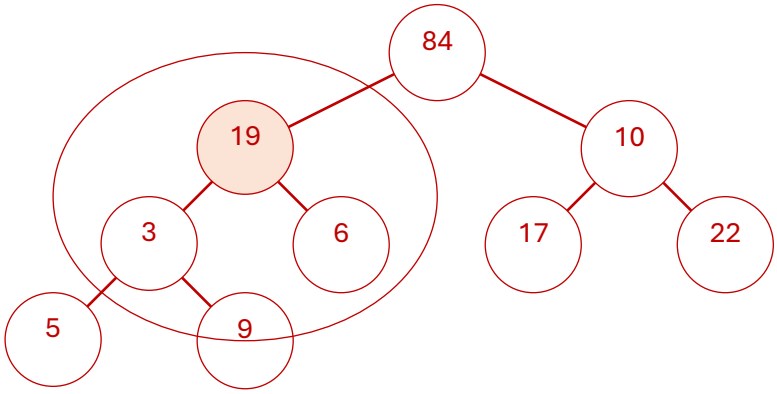
Starting from the parent of the last node (nlast), we reconstruct the heap using the convertToMinHeap operation.

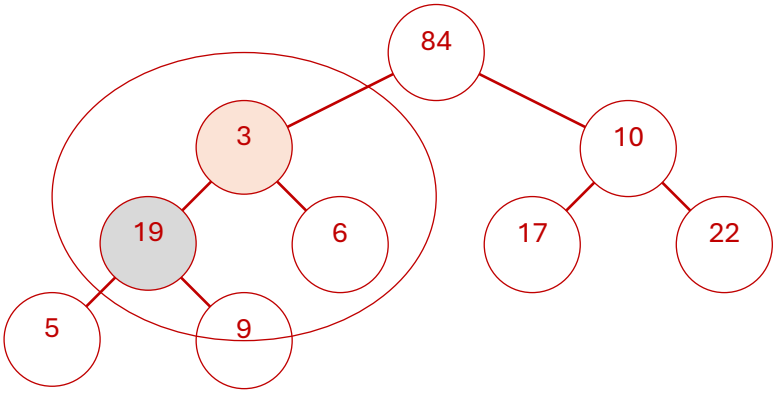


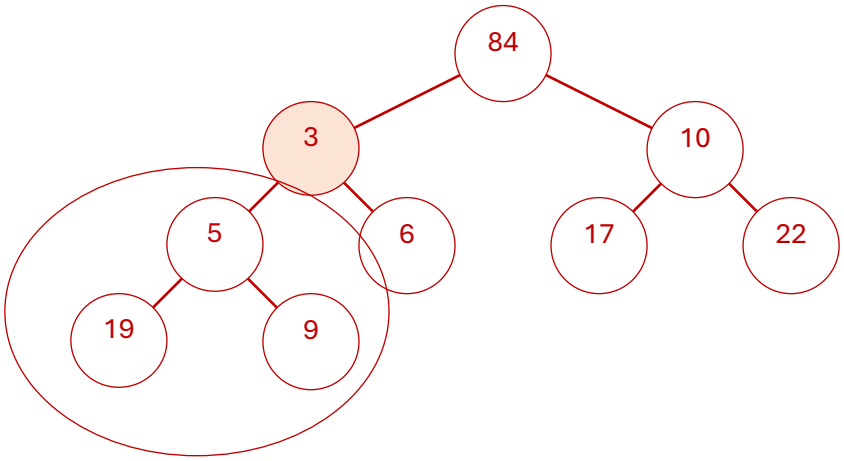


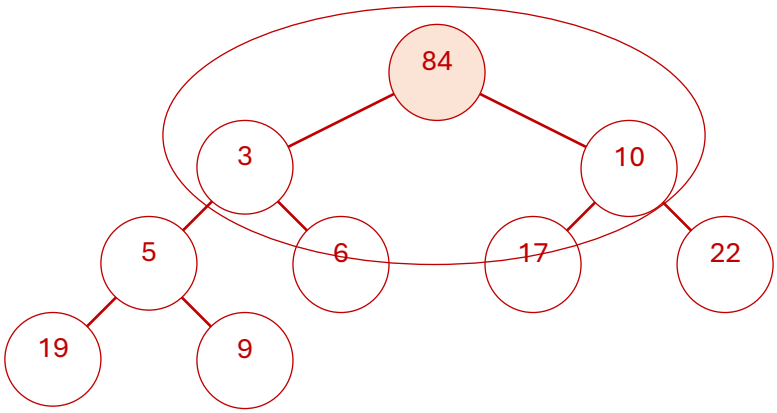


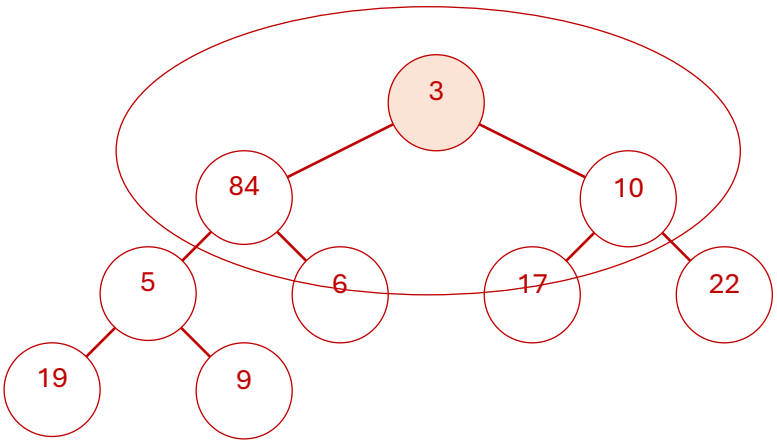


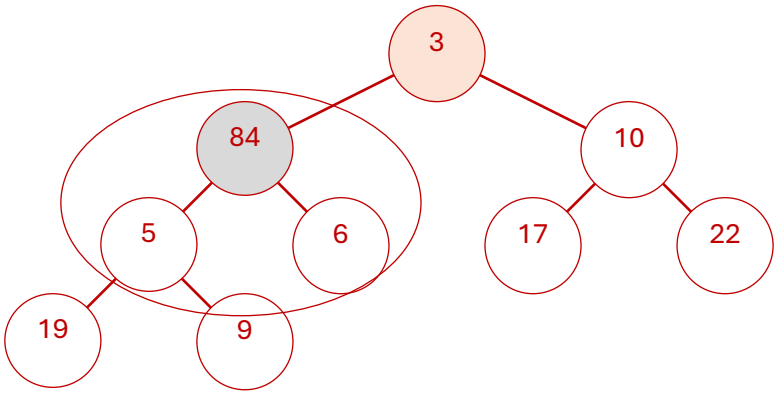


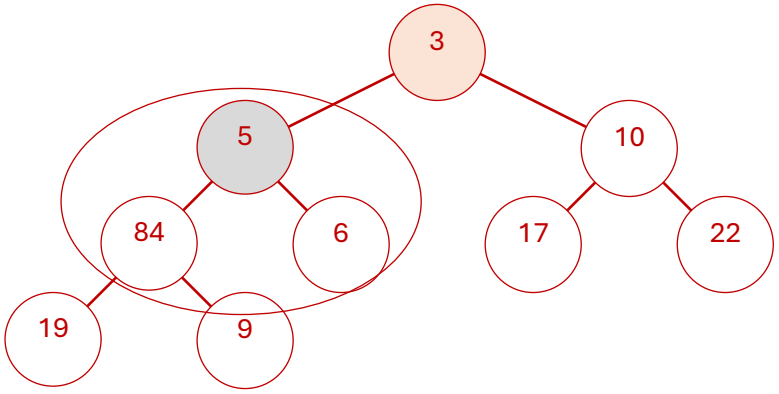


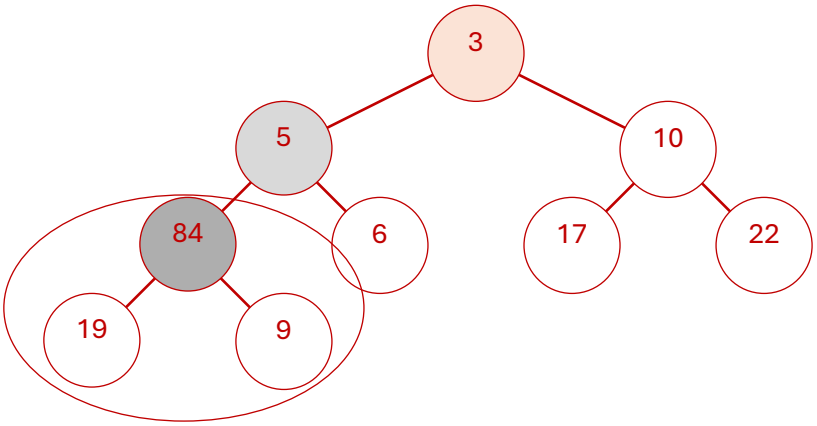


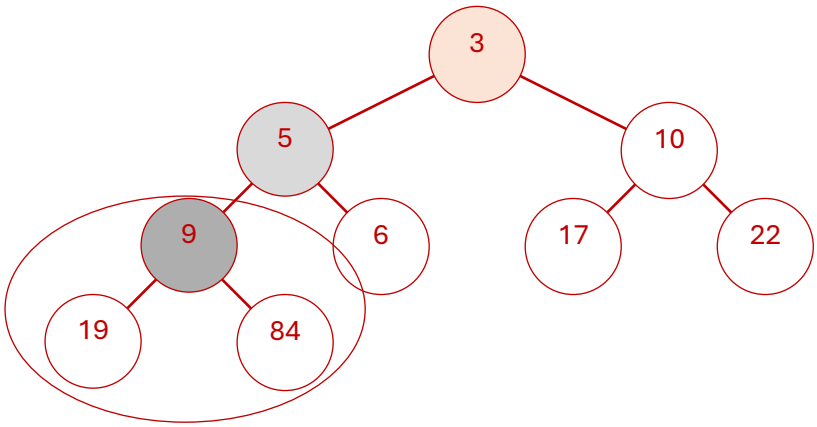




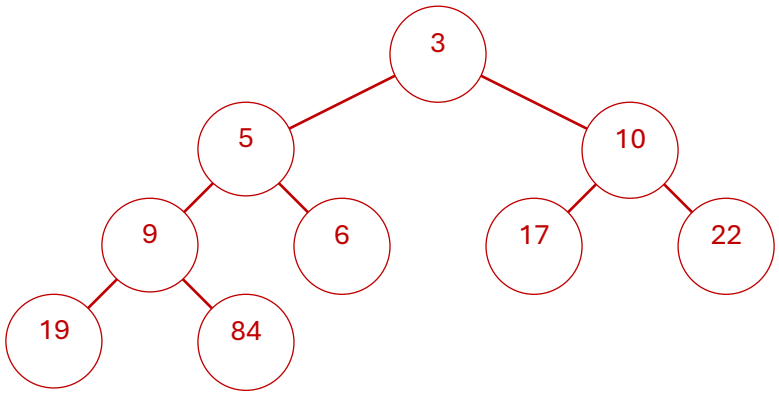








Reconstruction process completed. The heap is now a minimum heap.



Part B: Programming Question (30 Marks)

Assignment Brief: Logistics Hub Distribution Simulation

1. Overview

Your task is to investigate the efficiency of a two-stage automated logistics hub. Packages arrive at a **Sorting Station** (Primary) and, once processed, move to a **Loading Dock** (Secondary) for final dispatch. You must simulate this flow to determine bottlenecks and server efficiency.

2. Implementation Constraints

- **Language:** Java, C++, or Python.

- **No Library Queues:** You are **strictly prohibited** from using built-in queue classes (e.g., java.util.LinkedList, std::queue, or collections.deque).
- **Custom Implementation:** You must write your own Queue class/struct using arrays or linked nodes.

3. System Logic

1. **Stage 1 (Sorting):** All packages enter a single "Arrival Queue." When a Sorting Server becomes available, the package is processed.
2. **Stage 2 (Loading):** Immediately after being sorted, the package enters a "Loading Queue." When a Loading Dock Server becomes available, the package is loaded for dispatch.
3. **Non-Blocking Flow:** A package cannot start Loading until its Sorting time is complete.

4. Input Format

The program should read from a data file provided via command line.

- **Line 1:** Two integers: S (Number of Sorting Servers) and L (Number of Loading Dock Servers).
- **Subsequent Lines:** Each line represents one package: Arrival_Time, Sorting_Time, and Loading_Time.
- **Termination:** A dummy record 0 0 0 signals the end of the input.

Example Data (logistics.dat):

2 1
1 5 10
2 3 5
5 2 2
0 0 0

(Translation: 2 Sorting Servers, 1 Loading Dock. Package 1 arrives at minute 1, takes 5 mins to sort and 10 mins to load.)

5. Required Output

Upon completion (when the last package has left the Loading Dock), output the following statistics:

- **Throughput:** Total number of packages processed.
- **Final Clock Time:** The time the last package cleared the system.
- **Processing Metrics:**
  - Average stay time in the hub (Arrival to Departure).
  - Average wait time in the Sorting Queue.
  - Average wait time in the Loading Queue.
- **Queue Analysis:**
  - Maximum length reached by the Sorting Queue.
  - Maximum length reached by the Loading Queue.

- **Server Efficiency:**
  - Total idle time for each individual Sorting Server.
  - Total idle time for each individual Loading Dock Server.

Note: There will be no sample solution provided for programming question.

Notes on Submission

1. This assignment is due on **Saturday, 14 February 2026 by 9:00 pm (21:00 hours) Singapore time**.
2. Submission must be made via **Moodle**. Ensure all files are combined into a single **ZIP file** without subdirectories.
3. Clearly label all components of your submission.
4. Late submissions will incur a **5% deduction per day**, including weekends.
5. Submissions more than **seven** days late will not be marked unless an extension is granted.
6. Extensions must be applied for via **SOLS** before the assignment deadline.
7. Plagiarism is treated seriously. Instances of plagiarism may result in a zero mark for the assignment.

Submit the file **SolutionA1.zip** through Moodle in the following way:

1) Access Moodle at **http://moodle.uowplatform.edu.au/**
2) To login use a Login link located in the right upper corner the Web page or in the middle of the bottom of the Web page
3) When successfully logged in, select a site **CSCI203 (SP126) Algorithms and Data Structures**
4) Scroll down to a section **Submissions of Assignments**
5) Click at **Submit** your Assignment 1 here link.
6) Click at a button **Add Submission**
7) Move the files created into an area provided in Moodle. You can drag and drop files here to add them. You can also use a link *Add…*
8) Click at a button **Save** changes,
9) Click at check box to confirm authorship of a submission,
10) When you are satisfied, remember to click at a button **Submit assignment**.

A policy regarding late submissions is included in the subject outline. Only one submission per student is accepted.

Assignment 1 is an individual assignment, and it is expected that all its tasks will be solved individually without any cooperation with the other students. Plagiarism is treated seriously. Students involved will likely receive zero. If you have any doubts, questions, etc. please consult your lecturer or tutor during lab classes or over e-mail.

End of specification

If you have any questions or require clarification, please contact your lecturer as early as possible. Good luck!